

Neural Networks and Deep Learning

23CSE473

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Exercise 2: Understanding Loss Functions (MSE vs BCE)

Objective

Compare Mean Squared Error (MSE) and Binary Cross Entropy (BCE) for binary classification.

Task 11: Compute the Mean Squared Error (MSE) Manually

Mean Squared Error (MSE) is a loss function that measures the average squared difference between the actual values and the predicted values. Although it is commonly used for regression problems, it can also be applied to binary classification to evaluate prediction errors.

Given:

True Labels:

$$y=[1,1,0,0]$$

Predicted Probabilities:

$$\hat{y} = [0.95, 0.80, 0.30, 0.10]$$

The Mean Squared Error is calculated using

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Substituting the given values,

$$\begin{aligned}MSE &= \frac{(1 - 0.95)^2 + (1 - 0.80)^2 + (0 - 0.30)^2 + (0 - 0.10)^2}{4} \\&= \frac{(0.05)^2 + (0.20)^2 + (0.30)^2 + (0.10)^2}{4} \\&= \frac{0.0025 + 0.0400 + 0.0900 + 0.0100}{4} \\&= \frac{0.1425}{4} \\&\boxed{MSE = 0.035625}\end{aligned}$$

Analysis

The computed Mean Squared Error is 0.035625, indicating that the predicted probabilities are close to the actual labels. Since the prediction errors are relatively small, the average squared error is also low, reflecting good prediction performance.

Task 12: Compute Binary Cross Entropy (BCE) Manually

Binary Cross Entropy (BCE) is a loss function commonly used for binary classification problems. It measures the difference between the actual labels and the predicted probabilities by assigning a higher penalty to incorrect predictions. Compared to Mean Squared Error, BCE is more suitable for classification tasks because it provides more informative gradients during training.

Given:

True Labels:

$$y = [1, 1, 0, 0]$$

Predicted Probabilities:

$$\hat{y} = [0.95, 0.80, 0.30, 0.10]$$

The Binary Cross Entropy is calculated using

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Substituting the given values,

$$BCE = -\frac{1}{4} [\log(0.95) + \log(0.80) + \log(0.70) + \log(0.90)]$$

Using natural logarithm values,

$$\log(0.95) = -0.0513$$

$$\log(0.80) = -0.2231$$

$$\log(0.70) = -0.3567$$

$$\log(0.90) = -0.1054$$

$$\begin{aligned} BCE &= -\frac{-0.0513 - 0.2231 - 0.3567 - 0.1054}{4} \\ &= \frac{0.7365}{4} \end{aligned}$$

$$\boxed{BCE = 0.1841}$$

Analysis

The Binary Cross Entropy loss obtained is **0.1841**, indicating that the predicted probabilities are close to the actual labels. BCE assigns a larger penalty to predictions that deviate significantly from the correct class, making it more effective than Mean Squared Error for binary classification tasks.

Task 13: Repeat the Calculations for Predictions [0.60, 0.55, 0.45, 0.40]

The Mean Squared Error (MSE) and Binary Cross Entropy (BCE) are recalculated using a new set of predicted probabilities while keeping the true labels unchanged. This comparison helps in understanding how both loss functions respond when the model predictions become less confident.

Given:

True Labels:

$$y = [1, 1, 0, 0]$$

Predicted Probabilities:

$$\hat{y} = [0.60, 0.55, 0.45, 0.40]$$

(a) Mean Squared Error (MSE)

The Mean Squared Error is calculated using

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Substituting the given values,

$$\begin{aligned} MSE &= \frac{(1 - 0.60)^2 + (1 - 0.55)^2 + (0 - 0.45)^2 + (0 - 0.40)^2}{4} \\ &= \frac{(0.40)^2 + (0.45)^2 + (0.45)^2 + (0.40)^2}{4} \\ &= \frac{0.1600 + 0.2025 + 0.2025 + 0.1600}{4} \\ &= \frac{0.7250}{4} \\ &= \boxed{MSE = 0.18125} \end{aligned}$$

(b) Binary Cross Entropy (BCE)

The Binary Cross Entropy is calculated using

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Substituting the given values

$$BCE = -\frac{1}{4} [\log(0.60) + \log(0.55) + \log(0.55) + \log(0.60)]$$

Using natural logarithm values,

$$\log(0.60) = -0.5108$$

$$\log(0.55) = -0.5978$$

$$\begin{aligned} BCE &= -\frac{-0.5108 - 0.5978 - 0.5978 - 0.5108}{4} \\ &= \frac{2.2172}{4} \end{aligned}$$

$$\boxed{BCE = 0.5543}$$

Analysis

Both the Mean Squared Error and Binary Cross Entropy values increased compared to the previous calculations because the predicted probabilities are less confident and farther from the true labels. The increase in BCE is more significant, showing that it penalizes uncertain predictions more strongly than MSE. This makes BCE more suitable for binary classification problems, where accurate probability estimation is important.

Task 14: Compare the Values

The computed Mean Squared Error (MSE) and Binary Cross Entropy (BCE) values for both sets of predicted probabilities are compared below.

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Comparison Table

Predicted Probabilities	MSE	BCE
[0.95, 0.80, 0.30, 0.10]	0.035625	0.1841
[0.60, 0.55, 0.45, 0.40]	0.18125	0.5543

Observation

- For the first set of predictions, both MSE and BCE are relatively low because the predicted probabilities are close to the true labels.
- For the second set of predictions, both loss values increase because the predictions are less accurate.
- The increase in **Binary Cross Entropy** is much larger than the increase in **Mean Squared Error**, indicating that BCE penalizes uncertain and incorrect predictions more strongly.

Analysis

The comparison shows that both loss functions reflect the quality of the predictions, but Binary Cross Entropy is more sensitive to prediction errors in binary classification. As the predicted probabilities move away from the true labels, BCE increases more significantly than MSE. This stronger penalty encourages the neural network to make more confident and accurate predictions during training.

Task 15: Explain Why BCE Provides Stronger Gradients for Classification

Binary Cross Entropy (BCE) is specifically designed for binary classification problems because it directly measures the difference between the predicted probabilities and the actual class labels. When the predicted probability is far from the correct value, BCE produces a larger error, resulting in stronger gradients during backpropagation. These stronger gradients enable the neural network to update its weights more effectively and learn faster.

In contrast, Mean Squared Error (MSE) measures the squared difference between the predicted value and the actual value. As the prediction approaches the correct value, the gradient produced by MSE becomes smaller. When sigmoid activation is used, this effect becomes more noticeable because the gradients can become very small, slowing down the learning process.

For example, if the true label is **1** but the model predicts **0.10**, Binary Cross Entropy assigns a much larger penalty than Mean Squared Error. Consequently, the network performs a larger weight update to correct the prediction. This allows BCE to converge faster and produce better classification performance.

Therefore, Binary Cross Entropy is generally preferred over Mean Squared Error for binary classification tasks because it provides stronger learning signals, faster convergence, and more reliable probability estimation.

Analysis

Binary Cross Entropy improves the learning process by generating larger gradients whenever the model makes incorrect or uncertain predictions. These stronger gradients help the optimizer update the network parameters more efficiently, resulting in faster convergence and better classification performance. For this reason, BCE is the standard loss function used in most binary classification problems.

Task 16: Train Two Identical Neural Networks

Explanation

To compare the performance of Mean Squared Error (MSE) and Binary Cross Entropy (BCE), two identical Multi-Layer Perceptron (MLP) models were trained using the **Banknote Authentication Dataset**. Both models use the same network architecture, optimizer, activation function, number of epochs, and training data. The only difference between the two models is the loss function. Model A uses Mean Squared Error (MSE), while Model B uses Binary Cross Entropy (BCE). This experimental setup allows a fair comparison of the effect of the two loss functions on binary classification.

CODE::

```
# Task 16: Comparing MSE and BCE using Banknote Authentication Dataset

import pandas as pd
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
|
# Reproducibility
np.random.seed(42)
tf.random.set_seed(42)

# Load dataset
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/00267/data_banknote_authentication.txt"

columns = [
    "Variance",
    "Skewness",
    "Curtosis",
    "Entropy",
    "Class"
]
```

```

    model.compile(
        optimizer='adam',
        loss=loss_function,
        metrics=['accuracy']
    )

    return model

# Model A (MSE)
model_mse = build_model('mean_squared_error')

history_mse = model_mse.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=16,
    validation_data=(X_test, y_test),
    verbose=0
)

# Model B (BCE)
model_bce = build_model('binary_crossentropy')

history_bce = model_bce.fit(
    X_train,
    y_train,
    epochs=50,

```

```

        batch_size=16,
        validation_data=(X_test, y_test),
        verbose=0
    )

# Model B (BCE)
model_bce = build_model('binary_crossentropy')

history_bce = model_bce.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=16,
    validation_data=(X_test, y_test),
    verbose=0
)

```

```

        batch_size=16,
        validation_data=(X_test, y_test),
        verbose=0
    )

    # Evaluate both models
    loss_mse, acc_mse = model_mse.evaluate(X_test, y_test, verbose=0)

    loss_bce, acc_bce = model_bce.evaluate(X_test, y_test, verbose=0)

    print("Model A (MSE)")
    print(["Test Accuracy:", round(acc_mse*100,2), "%"])
    print()

    print("Model B (BCE)")
    print(["Test Accuracy:", round(acc_bce*100,2), "%"])

```

OUTPUT:

```

... /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model A (MSE)
Test Accuracy: 98.55 %

Model B (BCE)
Test Accuracy: 99.64 %

```

Analysis

Both models achieved high classification accuracy on the Banknote Authentication dataset. However, the model trained using Binary Cross Entropy (BCE) obtained a slightly higher accuracy (99.64%) than the model trained using Mean Squared Error (MSE) (98.55%). This indicates that BCE is more suitable for binary classification tasks.

Task 17: Compare Training Loss, Accuracy, and Convergence Speed

Two identical neural networks were trained using different loss functions: Mean Squared Error (MSE) and Binary Cross Entropy (BCE). Their performance was compared based on training loss, test accuracy, and convergence speed to evaluate the suitability of each loss function for binary classification.

CODE:

```
# Task 17: Compare Training Loss

import matplotlib.pyplot as plt

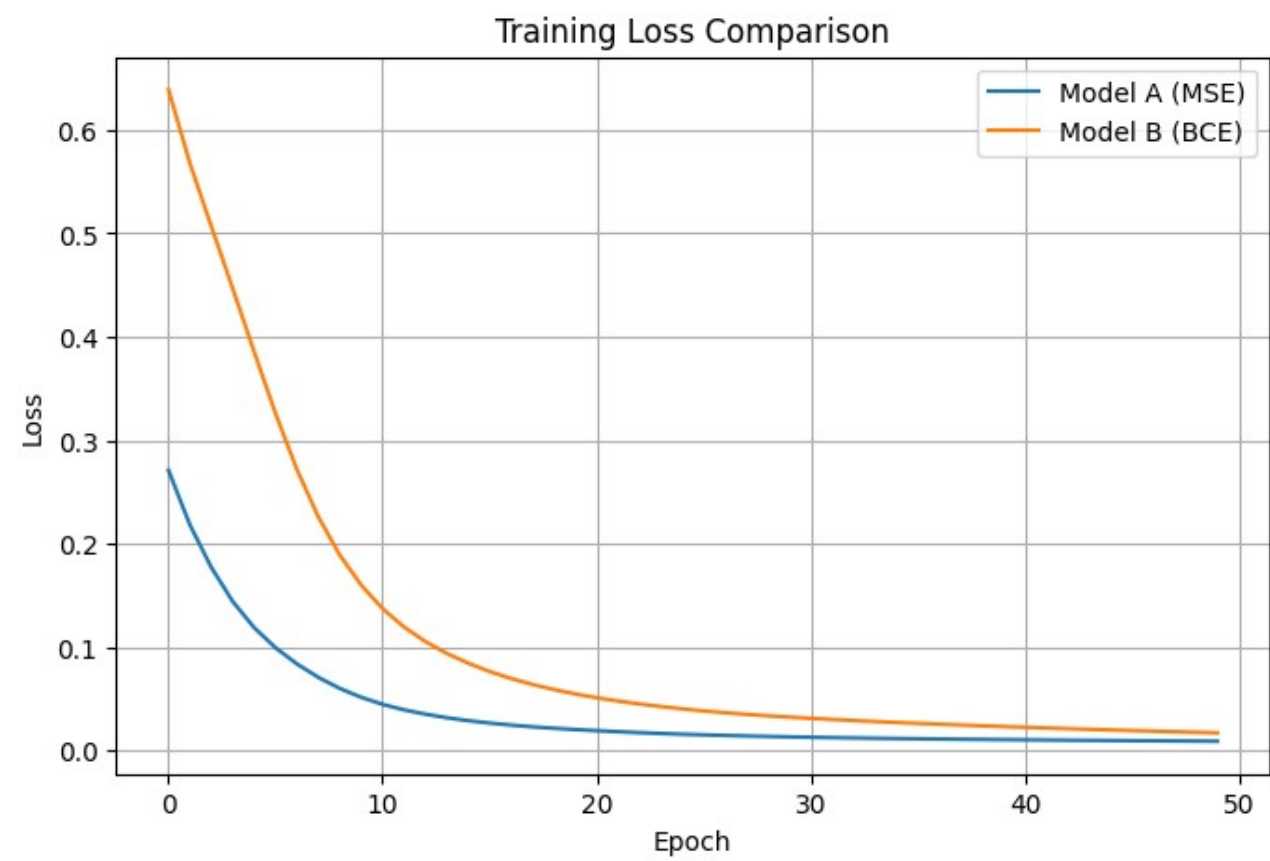
plt.figure(figsize=(8,5))

plt.plot(history_mse.history['loss'], label='Model A (MSE)')
plt.plot(history_bce.history['loss'], label='Model B (BCE)')

plt.title("Training Loss Comparison")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)

plt.show()
```

Output :



Comparison Table

Parameter	Model A (MSE)	Model B (BCE)
Loss Function	Mean Squared Error	Binary Cross Entropy
Test Accuracy	98.55%	99.64%
Convergence	Slower	Faster
Suitable For	Regression / Basic Classification	Binary Classification

Analysis

Both models showed a steady decrease in training loss as the number of epochs increased, indicating successful learning. The model trained with Binary Cross Entropy achieved a slightly higher test accuracy (99.64%) than the model trained with Mean Squared Error (98.55%). These results suggest that Binary Cross Entropy is more effective for binary classification tasks because it is specifically designed for probability-based predictions.

Conclusion

This experiment compared Mean Squared Error (MSE) and Binary Cross Entropy (BCE) for binary classification. Manual calculations showed that BCE penalizes incorrect predictions more strongly than MSE. Experimental results using two identical neural networks on the Banknote Authentication dataset further demonstrated that the model trained with BCE achieved slightly higher accuracy than the model trained with MSE. Therefore, Binary Cross Entropy is generally a more suitable loss function for binary classification problems.